

# Mie holography and optical trapping to study colloid-colloid interaction

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## 1 Experimental Method

The method used to monitor the interaction between two brownian particles is known as *Mie Holography* [1, 2]. This method relies on the holograms produced by interferences between the incident light illuminating a particle, and the light scattered by the latter (see Fig. 1a). The interferences patterns are extracted as intensity profiles (see Fig. 1b) and fitted based on the Lorenz-Mie theory to recover the tridimensional position of the particle, for every frame composing the video.

The sample consists of two glass slides (thickness  $e = 0.13 - 0.17$  mm and refractive index  $n_g = 1.518$ ) separated by paraffin bands. The space  $H$  between the two glass slides, set by the thickness of the paraffin bands, is about  $H = 250$   $\mu\text{m}$ . The Brownian microparticles are made of polystyrene, manufactured by Polyscience. They are negatively surface-charged and have a radius  $a_p = 1.5$   $\mu\text{m}$ , for a density of  $\rho_p = 1050$  g/cm<sup>3</sup>. Their motion is recorded through a x100 oil-immersion Olympus objective (NA = 1.45), configured for TIRFM, using a Basler Camera, with a field of view of  $70 \times 70$   $\mu\text{m}$  in the  $xOy$  plane. Along  $z$ , the depth of view  $h$  is about 100  $\mu\text{m}$ . The motion of the particles is recorded at a frequency  $f = \frac{1}{dt} = 200$  Hz, leaving a lag time of  $dt = 5$  ms between two frames.

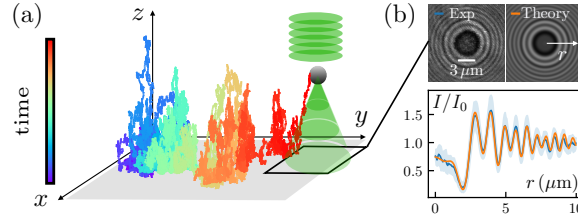


Figure 1: Mie Holography method used to probe the motion of a colloid 1.5  $\mu\text{m}$ -radius particles in water, illuminated by a laser at 532 nm at a laser power of 4 mW.

In order to enforce the hydrodynamic interaction between two particles, we set a *optical trap* [3], enabling the translation of a particle from one point to an other (see Fig. 2). The method consist in finding a diffusing particle, move it as close as possible to another one, and take a video with a high frame rate to have as much data as possible, in the region where the particle-particle interaction is omnipresent.

## References

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- [2] M. Lavaud, T. Salez, Y. Louyer, and Y. Amarouchene, “Stochastic inference of surface-induced effects using brownian motion,” *Physical Review Research*, vol. 3, no. 3, p. L032011, 2021.
- [3] A. Ashkin, “Forces of a single-beam gradient laser trap on a dielectric sphere in the ray optics regime,” *Biophysical journal*, vol. 61, no. 2, pp. 569–582, 1992.

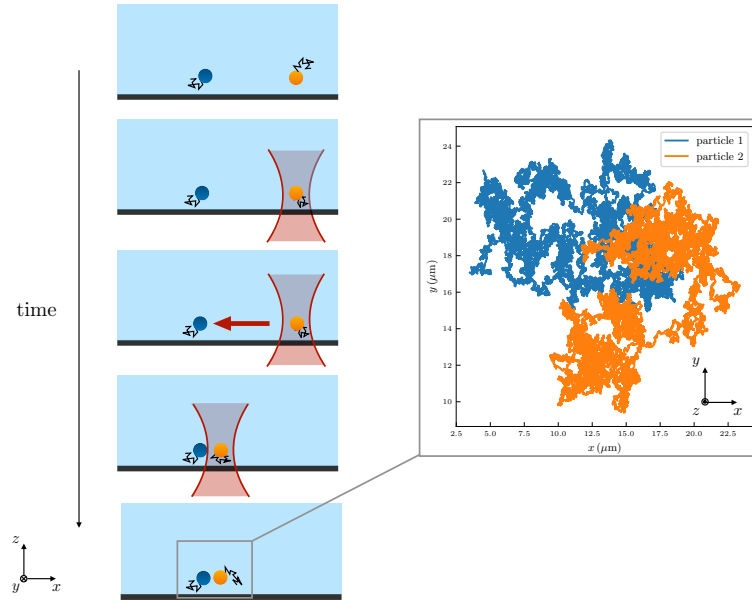


Figure 2: Optical trapping used to enforce the colloid-colloid interaction. The trap is set by a laser at 660 nm at a laser power of 50 mW.